

Exploring Multi-Armed Bandits in Recommendation System

Project Report

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Problem Statement

Modern recommendation systems must adapt in real time to individual users’ preferences, evolving inventories, and shifting behavioral patterns. Traditional offline models struggle with the “cold-start” problem, nonstationary user interests, and the need to continuously balance exploration of new items against exploitation of known favorites.

In this project, we address these challenges by formulating product recommendation as a **contextual multi-armed bandit** problem. We evaluate and compare a suite of bandit algorithms—including LinUCB (ridge-regularized), ϵ -greedy (ridge and logistic variants), Thompson Sampling with online logistic regression, and their non-contextual counterparts—on two real-world logged datasets: MovieLens 1M (50 most-rated movies, 109,804 interactions) and Zozo/men (33 men’s fashion items, 440,435 impressions).

Our goal is to maximize cumulative click-through and conversion rates by selecting, at each user interaction, the arm (product) with the highest expected reward while still exploring under-seen items. We measure performance via cumulative reward, worst-case regret (vs. a random policy), and runtime, using the replay method for unbiased offline evaluation. This comparative study lays the groundwork for deploying adaptive, data-driven recommenders that learn from live implicit feedback.

Literature Review

After surveying the literature on multi-armed bandit (MAB) techniques in recommender systems, I selected two seminal studies for detailed examination and implemented their methodologies. Below is a concise review of these works.

- **Xiang (2021).** *Adaptively Optimize Content Recommendation Using Multi Armed Bandit Algorithms in E-commerce.*

Evaluation of ϵ -greedy, Thompson Sampling and UCB1 under non-stationary and delayed-feedback conditions; introduces a batch-update framework and demonstrates up to **6.13%** CTR and **16.1%** CVR improvements in a live App homepage A/B test.

- **Gangan (2021).** *Survey of Multi-Armed Bandit Algorithms applied to Recommendation Systems.*

A concise overview of bandit methods (ϵ -greedy, UCB, Thompson Sampling, EXP3), their contextual extensions and advanced variants (multi-play, multi-objective, clustering, collaborative filtering), with industry case studies (Spotify, Netflix, Yahoo) and empirical experiments on two datasets, emphasizing the importance of parameter tuning.

Areas (Gap) for Future Exploration

The paper comprehensively catalogs both contextual and non-contextual bandit algorithms for recommendation, but it stops short of addressing several **practical “productization” challenges** that remain critical for real-world deployment.:

- **Context (feature) engineering**

Determining *which* and *how* to encode user- and item-side features into the context vector is left as an open problem. Poorly chosen or noisy features can severely degrade online learning performance.

- **Dynamic action (arm) management**

In live systems, new items (arms) must be **added on the fly** without restarting the learner. The survey does not cover algorithms or architectures for seamlessly incorporating previously unseen arms into a running model.

However the main **aim** of this project is to **explore and compare various MAB approaches**

Detailed Problem Statement

Modern online services (e.g., Netflix, Spotify, Amazon) rely on recommender systems to personalize content and maximize user engagement. Conventional offline models—such as collaborative filtering and matrix factorization—suffer from three key limitations:

- **Cold-start:** New users/items lack interaction history, hampering accurate recommendations.
- **Nonstationarity:** User tastes and item popularities evolve over time, yet large models are expensive to retrain continuously.
- **Exploration–exploitation trade-off:** Static predictors cannot systematically explore underseen items without sacrificing short-term performance.

To overcome these challenges, we cast recommendation as a *contextual multi-armed bandit* problem. At each interaction round $t = 1, \dots, T$:

1. Observe context $\mathbf{x}_t \in \mathbb{R}^d$ encoding user features (e.g., demographics, past clicks) and item metadata (e.g., genre, price).
2. Select an arm $a_t \in \{1, \dots, K\}$ corresponding to one candidate item.
3. Receive immediate reward $r_{t,a_t} \in [0, 1]$ (e.g., click or dwell time).

The objective is to learn a policy $\pi(a \mid \mathbf{x})$ that maximizes cumulative reward

$$R_T = \sum_{t=1}^T r_{t,a_t}$$

or equivalently minimizes regret

$$\mathcal{R}_T = \sum_{t=1}^T (r_t^* - r_{t,a_t}), \quad r_t^* = \max_{a \in [K]} \mathbb{E}[r_{t,a} \mid \mathbf{x}_t].$$

Experiments employ *unbiased offline replay* on two logged **datasets**:

- **MovieLens 1M**: 50 most-rated movies, 109,804 interactions.
- **Zozo/men**: 33 men’s fashion items, 440,435 impressions.

Metrics include cumulative reward and worst-case regret versus a random policy.

Goal. Provide a reproducible, comparative study of bandit algorithms in real-world recommendation settings, yielding practical guidance for deploying adaptive, exploration-aware recommenders that learn from live implicit feedback.

Datasets

MovieLens 1M

The MovieLens 1M dataset (Harper & Konstan, 2015) comprises three files:

- **Ratings (ratings.csv)**: 1,000,209 records of user–movie ratings.
 - *Columns*: `UserID` (int), `MovieID` (int), `Rating` (1–5 float), `Timestamp` (UNIX epoch).
 - *Unique users*: 6,040; *unique movies*: 3,706.
- **Movie metadata (movies.csv)**: 3,883 movie entries.
 - *Columns*: `MovieID` (int), `Title` (string), `Genres` (pipe-separated).
- **User demographics (users.csv)**: 6,040 user profiles.
 - *Columns*: `UserID` (int), `Gender` (M/F), `Age` (int), `Occupation` (int code), `Zip-code` (string).

Zozo Men’s Fashion Impressions

This dataset contains logged impressions on 34 men’s fashion items:

- **Interactions (men.csv):** 10,000 impression events.
 - *Columns:* timestamp (UNIX time), item_id (int), position (int), click (0/1), propensity_score (float), user_feature_0–user_feature_3, user-item_affinity_0–user-item_affinity_33.
 - *Unique items:* 34.
- **Item context (item_context.csv):** 34 item records.
 - *Columns:* item_id (int), item_feature_0–item_feature_3.

Algorithms and Evaluation

We compare both contextual and non-contextual bandits:

- **LinUCB:** ridge-regularized linear UCB.
- **ϵ -Greedy:** constant/decreasing exploration with ridge/logistic regression.
- **Thompson Sampling:** Bayesian sampling via online logistic regression.
- **UCB, standard ϵ -Greedy.:**

LinUCB with Disjoint Linear Models

Algorithm 1 LinUCB with Disjoint Linear Models

Require: Exploration parameter $\alpha > 0$, number of arms K , feature dimension d

```
1: for  $a = 1$  to  $K$  do
2:    $A_a \leftarrow I_d$ 
3:    $b_a \leftarrow \mathbf{0}_d$ 
4: end for
5: for  $t = 1, 2, \dots, T$  do
6:   Observe  $\{x_{a,t} \in \mathbb{R}^d\}_{a=1}^K$ 
7:   for each arm  $a = 1, \dots, K$  do
8:      $\hat{\theta}_a \leftarrow A_a^{-1} b_a$ 
9:      $p_{a,t} \leftarrow x_{a,t}^\top \hat{\theta}_a + \alpha \sqrt{x_{a,t}^\top A_a^{-1} x_{a,t}}$ 
10:  end for
11:  Select  $a_t \leftarrow \arg \max_a p_{a,t}$ 
12:  Receive reward  $r_t \in [0, 1]$ 
13:   $A_{a_t} \leftarrow A_{a_t} + x_{a_t,t} x_{a_t,t}^\top$ 
14:   $b_{a_t} \leftarrow b_{a_t} + r_t x_{a_t,t}$ 
15: end for
```

Epsilon-Greedy with Ridge Regression

Algorithm 2 Epsilon-Greedy with Ridge Regression

Require: Exploration rate $\epsilon \in [0, 1]$, number of arms K , feature dimension d , ridge regularization

$\lambda > 0$

- 1: **for** $a = 1, \dots, K$ **do**
- 2: $A_a \leftarrow \lambda I_d$ $\{d \times d$ ridge covariance matrix $\}$
- 3: $b_a \leftarrow \mathbf{0}_d$ $\{d$ -dim. response vector $\}$
- 4: **end for**
- 5: **for** $t = 1, \dots, T$ **do**
- 6: Observe feature vectors $\{x_{a,t} \in \mathbb{R}^d\}_{a=1}^K$
- 7: Draw $u \sim \text{Uniform}(0, 1)$
- 8: **if** $u < \epsilon$ **then**
- 9: Choose arm $a_t \sim \text{Uniform}\{1, \dots, K\}$
- 10: **else**
- 11: **for** $a = 1, \dots, K$ **do**
- 12: $\hat{r}_a \leftarrow x_{a,t}^\top A_a^{-1} b_a$
- 13: **end for**
- 14: Select arm $a_t \leftarrow \arg \max_a \hat{r}_a$
- 15: **end if**
- 16: Receive reward $r_t \in [0, 1]$
- 17: // **Update ridge regression state**
- 18: $A_{a_t} \leftarrow A_{a_t} + x_{a_t,t} x_{a_t,t}^\top$
- 19: $b_{a_t} \leftarrow b_{a_t} + r_t x_{a_t,t}$
- 20: **end for**

Epsilon-Greedy with Logistic Regression

Algorithm 3 Epsilon-Greedy with Logistic Regression

Require: Exploration rate $\epsilon \in [0, 1]$, number of arms K , feature dimension d , learning rate $\eta > 0$

```

1: for  $a = 1, \dots, K$  do
2:    $\theta_a \leftarrow \mathbf{0}_d$  {Initialize logistic weights}
3: end for
4: for  $t = 1, \dots, T$  do
5:   Observe feature vectors  $\{x_{a,t} \in \mathbb{R}^d\}_{a=1}^K$ 
6:   Draw  $u \sim \text{Uniform}(0, 1)$ 
7:   if  $u < \epsilon$  then
8:     Select arm  $a_t \sim \text{Uniform}\{1, \dots, K\}$ 
9:   else
10:    for  $a = 1, \dots, K$  do
11:       $\hat{p}_a \leftarrow \sigma(x_{a,t}^\top \theta_a)$   $\{\sigma(z) = 1/(1 + e^{-z})\}$ 
12:    end for
13:    Select arm  $a_t \leftarrow \arg \max_a \hat{p}_a$ 
14:  end if
15:  Receive binary reward  $r_t \in \{0, 1\}$ 
16:  // Online logistic regression update
17:   $\theta_{a_t} \leftarrow \theta_{a_t} + \eta (r_t - \sigma(x_{a_t,t}^\top \theta_{a_t})) x_{a_t,t}$ 
18: end for
```

Epsilon-Greedy (Context-Free)

Algorithm 4 Epsilon-Greedy (Context-Free)

Require: Exploration rate $\epsilon \in [0, 1]$, number of arms K , total rounds T

```

1: Initialize counts  $N(a) \leftarrow 0$  and estimates  $Q(a) \leftarrow 0$  for all  $a = 1, \dots, K$ 
2: for  $t = 1, \dots, T$  do
3:   Draw  $u \sim \text{Uniform}(0, 1)$ 
4:   if  $u < \epsilon$  then
5:     Select arm  $a_t \sim \text{Uniform}\{1, \dots, K\}$ 
6:   else
7:     Select arm  $a_t \leftarrow \arg \max_a Q(a)$ 
8:   end if
9:   Receive reward  $r_t \in [0, 1]$ 
10:   $N(a_t) \leftarrow N(a_t) + 1$ 
11:   $Q(a_t) \leftarrow Q(a_t) + \frac{1}{N(a_t)} (r_t - Q(a_t))$ 
12: end for
```

Thompson Sampling with Online Logistic Regression

Algorithm 5 Thompson Sampling with Online Logistic Regression

Require: Prior precision $\lambda > 0$, number of arms K , feature dimension d

```

1: for  $a = 1, \dots, K$  do
2:    $A_a \leftarrow \lambda I_d$                                      {Approx. posterior precision}
3:    $b_a \leftarrow \mathbf{0}_d$                                 {Precision-times-mean}
4:    $m_a \leftarrow A_a^{-1} b_a$                                {Posterior mean}
5: end for
6: for  $t = 1, \dots, T$  do
7:   Observe context vectors  $\{x_{a,t} \in \mathbb{R}^d\}_{a=1}^K$ 
8:   for  $a = 1, \dots, K$  do
9:     Draw sample  $\theta_a \sim \mathcal{N}(m_a, A_a^{-1})$ 
10:    Compute score  $p_{a,t} \leftarrow \sigma(x_{a,t}^\top \theta_a)$ 
11:  end for
12:  Select arm  $a_t \leftarrow \arg \max_a p_{a,t}$ 
13:  Receive binary reward  $r_t \in \{0, 1\}$ 
14:  // Laplace update (one-step)
15:   $\tilde{p} \leftarrow \sigma(x_{a_t,t}^\top m_{a_t})$ 
16:   $s \leftarrow \tilde{p} (1 - \tilde{p})$ 
17:   $A_{a_t} \leftarrow A_{a_t} + s x_{a_t,t} x_{a_t,t}^\top$ 
18:   $b_{a_t} \leftarrow b_{a_t} + (r_t - \tilde{p}) x_{a_t,t}$ 
19:   $m_{a_t} \leftarrow A_{a_t}^{-1} b_{a_t}$ 
20: end for

```

Random Policy

Algorithm 6 Random Policy

Require: Number of arms K , total rounds T

```

1: for  $t = 1, \dots, T$  do
2:   Select arm  $a_t \sim \text{Uniform}\{1, \dots, K\}$ 
3:   Receive reward  $r_t$ 
4: end for

```

Thompson Sampling (Context-Free)

Algorithm 7 Thompson Sampling (Context-Free)

Require: Number of arms K , total rounds T

```
1: Initialize successes  $S(a) \leftarrow 0$  and failures  $F(a) \leftarrow 0$  for all  $a = 1, \dots, K$ 
2: for  $t = 1, \dots, T$  do
3:   for  $a = 1, \dots, K$  do
4:     Sample  $\theta_a \sim \text{Beta}(S(a) + 1, F(a) + 1)$ 
5:   end for
6:   Select arm  $a_t \leftarrow \arg \max_a \theta_a$ 
7:   Receive binary reward  $r_t \in \{0, 1\}$ 
8:   if  $r_t = 1$  then
9:      $S(a_t) \leftarrow S(a_t) + 1$ 
10:  else
11:     $F(a_t) \leftarrow F(a_t) + 1$ 
12:  end if
13: end for
```

UCB1 (Context-Free)

Algorithm 8 UCB1 (Context-Free)

Require: Number of arms K , total rounds T

```
1: Initialize counts  $N(a) \leftarrow 0$ , value estimates  $Q(a) \leftarrow 0$  for  $a = 1, \dots, K$ 
2: for  $a = 1, \dots, K$  do
3:   Play arm  $a$ , receive reward  $r$ ,
4:    $N(a) \leftarrow 1$ ,  $Q(a) \leftarrow r$ 
5: end for
6: for  $t = K + 1, \dots, T$  do
7:   for  $a = 1, \dots, K$  do
8:      $U(a) \leftarrow Q(a) + \sqrt{\frac{2 \ln t}{N(a)}}$ 
9:   end for
10:  Select arm  $a_t \leftarrow \arg \max_a U(a)$ 
11:  Receive reward  $r_t \in [0, 1]$ 
12:   $N(a_t) \leftarrow N(a_t) + 1$ 
13:   $Q(a_t) \leftarrow Q(a_t) + \frac{1}{N(a_t)}(r_t - Q(a_t))$ 
14: end for
```

Conclusion

This study evaluated multiple contextual and non-contextual bandit algorithms for recommendation tasks across two real-world datasets. Our findings demonstrate that contextual approaches generally outperform their non-contextual counterparts on the MovieLens dataset and Zozo fashion dataset.

Key insights from our experiments include:

- Contextual algorithms provide substantial gains over context-free methods when user and item features are informative.
- Future work should address the productization challenges identified earlier, particularly scalable feature engineering techniques and dynamic arm management strategies for handling new items. Additional research could also explore hybrid models combining bandit methods with collaborative filtering to better leverage historical interaction patterns while maintaining adaptivity